Statistical Modelling & Analysis (STAT2004)
Applied Class (Week 6)

We will go through Questions 3 & 4 together in the Applied Class. Try the other questions yourself during the week and discuss your results with your tutor next week.

1. Suppose that $\boldsymbol{z}$ and $\boldsymbol{u}$ are nonzero vectors satisfying

$$\boldsymbol{u}^\mathsf{T}\mathbf{1} = \boldsymbol{z}^\mathsf{T}\mathbf{1} = \boldsymbol{z}^\mathsf{T}\boldsymbol{u} = 0 \qquad \text{and} \qquad \boldsymbol{z}^\mathsf{T}\boldsymbol{z} = \boldsymbol{u}^\mathsf{T}\boldsymbol{u}.$$

For a small positive constant δ, define

$$\boldsymbol{x}_1 = \boldsymbol{z} + \delta\boldsymbol{u}, \qquad \boldsymbol{x}_2 = \boldsymbol{z} - \delta\boldsymbol{u}, \qquad \boldsymbol{y} = \boldsymbol{u}.$$

All regression models include an intercept.

(a) Show that the sample correlation between $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ is

$$\frac{\boldsymbol{x}_1^T\boldsymbol{x}_2}{\|\boldsymbol{x}_1\|\|\boldsymbol{x}_2\|} = \frac{1-\delta^2}{1+\delta^2},$$

Solution. Since $\boldsymbol{z}$ and $\boldsymbol{u}$ are orthogonal to $\mathbf{1}$, both $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ have sample mean zero. Put $S = \boldsymbol{z}^\mathsf{T}\boldsymbol{z} = \boldsymbol{u}^\mathsf{T}\boldsymbol{u}$. Using $\boldsymbol{z}^\mathsf{T}\boldsymbol{u} = 0$,

$$\begin{aligned}\boldsymbol{x}_1^\mathsf{T}\boldsymbol{x}_2 &= (\boldsymbol{z}+\delta\boldsymbol{u})^\mathsf{T}(\boldsymbol{z}-\delta\boldsymbol{u}) \\ &= \boldsymbol{z}^\mathsf{T}\boldsymbol{z} - \delta^2\boldsymbol{u}^\mathsf{T}\boldsymbol{u} = (1-\delta^2)S.\end{aligned}$$

Also,

$$\|\boldsymbol{x}_1\|^2 = \|\boldsymbol{x}_2\|^2 = \boldsymbol{z}^\mathsf{T}\boldsymbol{z} + \delta^2\boldsymbol{u}^\mathsf{T}\boldsymbol{u} = (1+\delta^2)S.$$

Therefore

$$\operatorname{cor}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{(1-\delta^2)S}{(1+\delta^2)S} = \frac{1-\delta^2}{1+\delta^2}.$$

(b) The coefficient of determination (R-squared) can be evaluated as

$$R^2 = \frac{\sum_{i=1}^n(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n(y_i - \bar{y})^2}.$$

Show that each of the simple regressions of $\boldsymbol{y}$ on $\boldsymbol{x}_1$ and of $\boldsymbol{y}$ on $\boldsymbol{x}_2$ has

$$R^2 = \frac{\delta^2}{1+\delta^2}.$$

Interpret the results from parts (a) and (b) when δ is small. In particular, comment on the correlation between the two covariates and on their individual explanatory power for $\boldsymbol{y}$.

Solution. All vectors are centred, so $\bar{y} = 0$ and the fitted intercept is zero. For the regression on $\boldsymbol{x}_1$,

$$\widehat{\beta}_1 = \frac{\boldsymbol{x}_1^\mathsf{T}\boldsymbol{y}}{\boldsymbol{x}_1^\mathsf{T}\boldsymbol{x}_1} = \frac{\delta S}{(1+\delta^2)S} = \frac{\delta}{1+\delta^2}.$$

Thus $\widehat{\boldsymbol{y}} = \widehat{\beta}_1\boldsymbol{x}_1$, and the numerator in the given definition is

$$\sum_{i=1}^n(\widehat{y}_i - \bar{y})^2 = \widehat{\boldsymbol{y}}^\mathsf{T}\widehat{\boldsymbol{y}} = \widehat{\beta}_1^2\boldsymbol{x}_1^\mathsf{T}\boldsymbol{x}_1 = \frac{\delta^2}{1+\delta^2}S.$$

The denominator is

$$\sum_{i=1}^n(y_i - \bar{y})^2 = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} = S.$$

Hence $R^2 = \delta^2/(1+\delta^2)$. For the regression on $\boldsymbol{x}_2$, the slope is $-\delta/(1+\delta^2)$, so the fitted sum of squares and R^2 are unchanged.
When δ is small, the correlation between the covariates is close to one, whereas the R^2 from either simple regression is close to zero. Thus the covariates are almost identical, but neither individually explains much of $\boldsymbol{y}$.

(c) Express $\boldsymbol{y}$ as a linear combination of $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Hence, determine the coefficients and the value of R^2 for the multiple regression of $\boldsymbol{y}$ on $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$. Use your results to explain why high correlation between two covariates does not, by itself, justify removing one of them from a regression model.
Solution. Since $\boldsymbol{x}_1 - \boldsymbol{x}_2 = 2\delta\boldsymbol{u}$,

$$\boldsymbol{y} = \frac{1}{2\delta}\boldsymbol{x}_1 - \frac{1}{2\delta}\boldsymbol{x}_2.$$

The intercept is zero, the slopes are $1/(2\delta)$ and $-1/(2\delta)$, and the fit is exact, so $R^2 = 1$. Although the covariates are highly correlated, their small difference contains all the information about the response. Removing either covariate would therefore destroy most of the explanatory power.

2. Consider a linear regression model with design matrix $\boldsymbol{X}$ of full column rank. Suppose that observation i has residual $e_i = 0$.

(a) Show that deleting observation i does not change the least-squares coefficient estimates, provided that the design matrix for the remaining observations has full column rank.
Solution. The full-data normal equations are

$$\sum_{j=1}^{n} \boldsymbol{x}_j(y_j - \boldsymbol{x}_j^\mathsf{T}\widehat{\boldsymbol{\beta}}) = \boldsymbol{0}.$$

The contribution from observation i is $\boldsymbol{x}_i e_i = \boldsymbol{0}$. Removing this term leaves

$$\sum_{j\neq i} \boldsymbol{x}_j(y_j - \boldsymbol{x}_j^\mathsf{T}\widehat{\boldsymbol{\beta}}) = \boldsymbol{0}.$$

Thus the full-data estimate satisfies the normal equations for the remaining observations. Since their design matrix has full column rank, the solution is unique, so the coefficient estimates are unchanged.

(b) What effect does the deletion have on the usual unbiased estimator of the error variance?
Solution. The residual sum of squares is unchanged because the deleted residual is zero and the estimates do not change. If the model has p coefficients, the estimate changes from

$$\widehat{\sigma}^2 = \frac{\text{RSS}}{n-p} \quad \text{to} \quad \widehat{\sigma}^2_{(-i)} = \frac{\text{RSS}}{n-p-1}.$$

It therefore increases when RSS > 0.

3. The `trees` dataset in R contains measurements from 31 felled black cherry trees. The variables are:

- `Girth`: the diameter of the tree, measured in inches;
- `Height`: the height of the tree, measured in feet;
- `Volume`: the volume of timber, measured in cubic feet.

Although the first variable is called `Girth`, it is a measurement of diameter rather than circumference.

(a) Fit the linear regression model

$$\texttt{Volume} = \beta_0 + \beta_1 \texttt{Girth} + \beta_2 \texttt{Height} + \varepsilon.$$

Produce appropriate diagnostic plots and assess the assumptions of the linear regression model. Comment on any evidence of an inadequate mean structure, nonconstant error variance, or unusual observations.

Solution. The following code fits the model, obtains the error standard deviation, draws the standard diagnostic plots, and calculates measures used to investigate unusual observations.

```
trees_lm <- lm(Volume ~ Girth + Height, data = trees)
coef(trees_lm) ## fitted regression coefficients
summary(trees_lm)$sigma ## estimate of sigma
par(mfrow = c(2, 2))
plot(trees_lm) ## standard diagnostic plots
rstudent(trees_lm) ## externally studentised residuals
hatvalues(trees_lm) ## leverages
cooks.distance(trees_lm) ## Cook's distances
```

The fitted model is

$$\widehat{\texttt{Volume}} = -57.99 + 4.708\,\texttt{Girth} + 0.339\,\texttt{Height},$$

with estimated errors $N(0, 3.882^2)$. The residual-versus-fitted plot shows curvature, and the residual spread tends to increase with the fitted value. This suggests an inadequate linear mean structure and possibly nonconstant variance. Tree 31 has the largest externally studentised residual (about 2.77), high leverage and Cook's distance about 0.61, so it is unusual and influential, although not an overwhelming outlier.

(b) One possible way of allowing for curvature is to add quadratic terms. Fit the model

$$\texttt{Volume} = \beta_0 + \beta_1 \texttt{Girth} + \beta_2 \texttt{Girth}^2 + \beta_3 \texttt{Height} + \beta_4 \texttt{Height}^2 + \varepsilon.$$

Compare its diagnostic plots with those from part (a). Does the addition of quadratic terms adequately address the problems identified in part (a)? Comment briefly on whether there is a scientific reason to expect timber volume to be an additive quadratic function of diameter and height.

Solution. The code below fits the quadratic model and produces its diagnostic plots for comparison with part (a).

```
trees_quad <- lm(Volume ~ Girth + I(Girth^2) + Height +
 I(Height^2), data = trees)
coef(trees_quad) ## fitted regression coefficients
par(mfrow = c(2, 2))
plot(trees_quad) ## diagnostic plots
```

The quadratic model removes much of the curvature seen in part (a), although some unequal spread and unusual observations remain. The squared height term contributes little, and the raw powers also produce strong numerical collinearity. More importantly, an additive quadratic model has no clear geometric justification: volume should depend on the product of cross-sectional area and height, not on separate additive powers of diameter and height.

(c) Approximating the trunk of a tree by a cylinder suggests that its volume should be approximately proportional to its cross-sectional area multiplied by its height. This gives the relationship

$$\texttt{Volume} \approx c\,\texttt{Girth}^2\texttt{Height},$$

where `Girth` denotes diameter. Use this relationship to motivate a linear regression model for log(`Volume`) in terms of log(`Girth`) and log(`Height`). Fit the resulting model and assess its diagnostic plots. Compare the diagnostics with those from parts (a) and (b). Compare the

estimated coefficients of log(`Girth`) and log(`Height`) with the geometrically suggested values 2 and 1, respectively.

Solution. The following code fits the log-linear model, estimates its error standard deviation, produces diagnostic plots, and checks for unusual observations.

```
trees_log <- lm(log(Volume) ~ log(Girth) + log(Height),
 data = trees)
coef(trees_log) ## fitted regression coefficients
summary(trees_log)$sigma ## estimate of sigma on log scale
par(mfrow = c(2, 2))
plot(trees_log) ## diagnostic plots
rstudent(trees_log) ## studentised residuals
cooks.distance(trees_log) ## Cook's distances
```

Taking logarithms of the geometric relationship motivates

$$\log(\texttt{Volume}) = \beta_0 + \beta_1 \log(\texttt{Girth}) + \beta_2 \log(\texttt{Height}) + \varepsilon.$$

The fitted model is

$$\widehat{\log(\texttt{Volume})} = -6.632 + 1.983 \log(\texttt{Girth}) + 1.117 \log(\texttt{Height}),$$

with estimated errors $N(0, 0.0814^2)$. The residual-versus-fitted plot has no important curvature or funnel shape and is preferable to those from parts (a) and (b). Tree 18 has the largest studentised residual, about -2.33, but there is no compelling outlier. The estimated exponents are close to the geometrically suggested values 2 and 1.

(d) Retaining log(`Girth`) and log(`Height`) as the covariates, consider the Box–Cox family of transformations of the response, defined by

$$g_\lambda(y) = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log(y), & \lambda = 0. \end{cases}$$

Use the `boxcox` function from the `MASS` package to construct a Box–Cox profile likelihood for a model of `Volume` with predictors log(`Girth`) and log(`Height`), that is

$$g_\lambda(\texttt{Volume}) = \beta_0 + \beta_1 \log(\texttt{Girth}) + \beta_2 \log(\texttt{Height}) + \varepsilon.$$

Is $\lambda = 1$, which is equivalent to using the untransformed response in a model containing an intercept, plausible? Is $\lambda = 0$, corresponding to a logarithmic transformation, plausible? Taking account of both the profile likelihood and the diagnostic plots, recommend a final model.

Solution. Here the model supplied to `boxcox` retains the logged covariates but leaves the response untransformed. The function then profiles over transformations of `Volume`.

```
library(MASS)
trees_bc <- lm(Volume ~ log(Girth) + log(Height),
 data = trees)
bc <- boxcox(trees_bc, lambda = seq(-0.5, 0.5,
 length = 101))
bc$x[which.max(bc$y)] ## lambda at profile maximum
```

The profile likelihood is maximised at about $\widehat{\lambda} = -0.07$, with an approximate 95% interval $(-0.24, 0.11)$. Thus $\lambda = 0$ is plausible, whereas $\lambda = 1$ is clearly not. Together with the good diagnostics and geometric interpretation from part (c), this supports the log-linear model in part (c).

4. The `sat` data in the `faraway` package contain state-level educational expenditure and SAT results for the 50 US states in 1994–95. Fit the model

$$\texttt{total} = \beta_0 + \beta_1 \texttt{expend} + \beta_2 \texttt{salary} + \beta_3 \texttt{ratio} + \beta_4 \texttt{takers} + \epsilon.$$

```
library(faraway)
?sat
sat_lm <- lm(total ~ expend + salary + ratio + takers, data = sat)
```

(a) Calculate the studentised residuals. Are there any observations that might reasonably be regarded as outliers? Identify any such states and comment on the strength of the evidence.
Solution. The code calculates and orders the externally studentised residuals, obtains the two-sided Bonferroni critical value, and identifies the state having the largest absolute residual.

```
sat_rstud <- rstudent(sat_lm)
sort(sat_rstud) ## inspect the residuals
n <- nrow(sat)
bonf_crit <- qt(1 - 0.05/(2*n), df = 44)
bonf_crit ## Bonferroni critical value
sat[which.max(abs(sat_rstud)), ]
```

West Virginia has the largest absolute studentised residual, approximately -3.12. Under the model, each externally studentised residual has a t distribution with 44 degrees of freedom. For an overall significance level of 0.05, the two-sided Bonferroni critical value is

$$t_{44,\,1-0.05/(2\times 50)} \approx 3.526.$$

Since $|-3.12| < 3.526$, no state is declared an outlier after the Bonferroni correction. West Virginia is the most unusual observation and warrants inspection, but the evidence is insufficient at the 5% familywise level.

(b) Calculate the variance inflation factors for the four covariates. Is there evidence of collinearity? If so, identify the covariates principally involved and explain what the corresponding variance inflation factors indicate about the precision of their estimated coefficients.
Solution. The first command calculates the VIFs. The correlation matrix is useful for identifying which covariates are principally responsible for the large values.

```
vif(sat_lm) ## variance inflation factors
cor(sat[c("expend", "salary", "ratio", "takers")])
 ## correlations among covariates
```

The VIFs are approximately

	expend	salary	ratio	takers
VIF	9.5	9.3	2.4	1.8

There is substantial collinearity involving `expend` and `salary`, which are strongly positively associated. Their coefficient variances are about nine times those arising from comparable orthogonal covariates, so their standard errors are inflated by factors of about three. This makes their separate partial effects difficult to estimate precisely.

(c) Fit a reduced model obtained by dropping `expend`. Compare the estimated coefficient and standard error for `salary` in the two models, and recalculate the variance inflation factors. Explain the changes you observe. Does dropping `expend` materially change your conclusions about `salary`?
Solution. The following code fits the reduced model, extracts the salary estimate and standard error from each model, and calculates the reduced-model VIFs.

```
sat_reduced <- update(sat_lm, .  ~ .  - expend)
coef(summary(sat_lm))["salary", c("Estimate", "Std.  Error")]
coef(summary(sat_reduced))["salary",
 c("Estimate", "Std.  Error")]
vif(sat_reduced) ## VIFs after dropping expend
```

In the full model, the `salary` coefficient is about 1.64 with standard error 2.39, so there is little evidence of a partial salary effect. After dropping `expend`, its coefficient is about 2.55 with standard error about 1.00. The reduced-model VIFs are approximately 1.66 for `salary`,

1.08 for `ratio`, and 1.74 for `takers`. Removing `expend` greatly reduces the collinearity and the standard error of the salary coefficient. Salary now appears positively associated with total SAT score, conditional on `ratio` and `takers`. This is not evidence that expenditure is irrelevant: the reduced coefficient combines salary's association with the response with variation previously attributed to expenditure.